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Natural Language Processing Lab 2

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In this Lab Demo we will cover:

- Part of speech (POS) Tagging
- Applications of POS Tagging
- Penn Treebank
- POS tagging implementation using NLTK



POS Tagging

- The process of assigning a part-of-speech or lexical class marker to each word in a sentence (and all sentences in a collection).

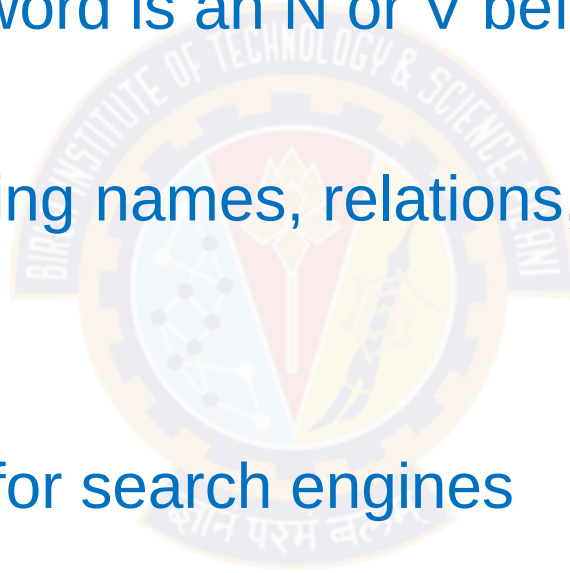
Input: the lead paint is unsafe

Output: the/Det lead/N paint/N is/V unsafe/Adj

- In corpus linguistics, **part-of-speech tagging** (POS tagging or PoS tagging or **POST**), is also called **grammatical tagging** or **word-category disambiguation**.

Why is POS Tagging Useful?

- First step of a vast number of practical tasks
- Helps in stemming/lemmatization
- Parsing: Need to know if a word is an N or V before you can parse
- Word sense disambiguation
- Information Extraction: Finding names, relations, etc.
- Machine Translation
- Text to speech conversion
- Pre-processing documents for search engines

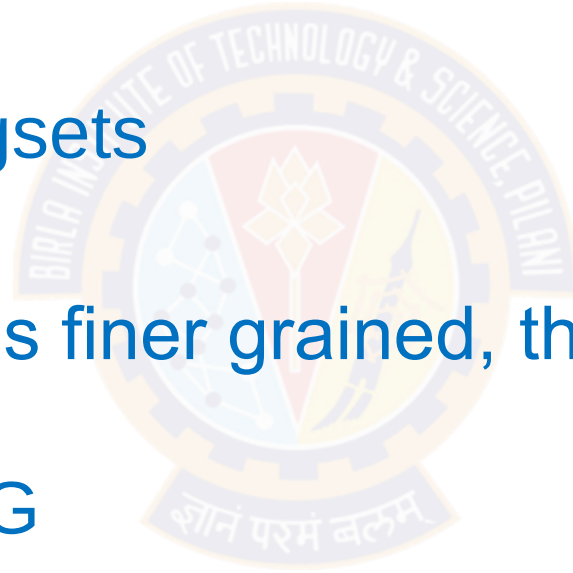


Part of speech

- Noun (N)- London, table, dog, teacher, pen, city, happiness, hope
- Determiner (DET) – a, an, the
- Verb (V)- go, speak, run, eat, play, live, walk, have, like, are, is
- Adjective(ADJ)- big, happy, green, young, fun, crazy, three
- Adverb(ADV)- slowly, quietly, very, always, never, too, well, tomorrow
- Preposition (P)- at, on, in, from, with, near, between, about, under
- Conjunction (CON)- and, or, but, because, so, yet, unless, since, if
- Pronoun(PRO)- I, you, we, they, he, she, it, me, us, them, him, her, this
- Interjection (INT)- Ouch! Wow! Great! Help! Oh! Hey! Hi!

POS Tagging

- There are so many parts of speech, potential distinctions we can draw
- To do POS tagging, we need to choose a standard set of tags to work with
- Could pick very coarse tagsets
 - N, V, Adj, Adv.
- More commonly used set is finer grained, the “Penn TreeBank tagset”, 45 tags
 - PRP\$, WRB, WP\$, VBG
- Even more fine-grained tagsets exist

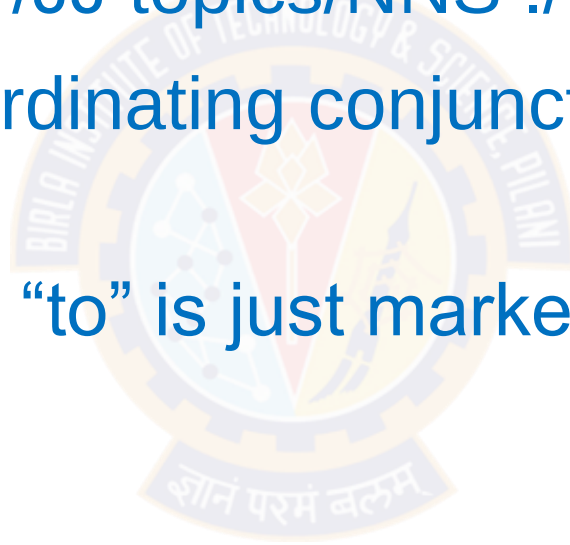


Penn TreeBank POS Tagset

Tag	Description	Example	Tag	Description	Example
CC	coordin. conjunction	<i>and, but, or</i>	SYM	symbol	<i>+, %, &</i>
CD	cardinal number	<i>one, two, three</i>	TO	“to”	<i>to</i>
DT	determiner	<i>a, the</i>	UH	interjection	<i>ah, oops</i>
EX	existential ‘there’	<i>there</i>	VB	verb, base form	<i>eat</i>
FW	foreign word	<i>mea culpa</i>	VBD	verb, past tense	<i>ate</i>
IN	preposition/sub-conj	<i>of, in, by</i>	VBG	verb, gerund	<i>eating</i>
JJ	adjective	<i>yellow</i>	VCN	verb, past participle	<i>eaten</i>
JJR	adj., comparative	<i>bigger</i>	VBP	verb, non-3sg pres	<i>eat</i>
JJS	adj., superlative	<i>wildest</i>	VBZ	verb, 3sg pres	<i>eats</i>
LS	list item marker	<i>1, 2, One</i>	WDT	wh-determiner	<i>which, that</i>
MD	modal	<i>can, should</i>	WP	wh-pronoun	<i>what, who</i>
NN	noun, sing. or mass	<i>llama</i>	WP\$	possessive wh-	<i>whose</i>
NNS	noun, plural	<i>llamas</i>	WRB	wh-adverb	<i>how, where</i>
NNP	proper noun, singular	<i>IBM</i>	\$	dollar sign	<i>\$</i>
NNPS	proper noun, plural	<i>Carolinas</i>	#	pound sign	<i>#</i>
PDT	predeterminer	<i>all, both</i>	“	left quote	<i>‘ or “</i>
POS	possessive ending	<i>’s</i>	”	right quote	<i>’ or ”</i>
PRP	personal pronoun	<i>I, you, he</i>	(	left parenthesis	<i>[, (, {, <</i>
PRP\$	possessive pronoun	<i>your, one’s</i>	)	right parenthesis	<i>],), }, ></i>
RB	adverb	<i>quickly, never</i>	,	comma	<i>,</i>
RBR	adverb, comparative	<i>faster</i>	.	sentence-final punc	<i>. ! ?</i>
RBS	adverb, superlative	<i>fastest</i>	:	mid-sentence punc	<i>: ; ... - -</i>
RP	particle	<i>up, off</i>			

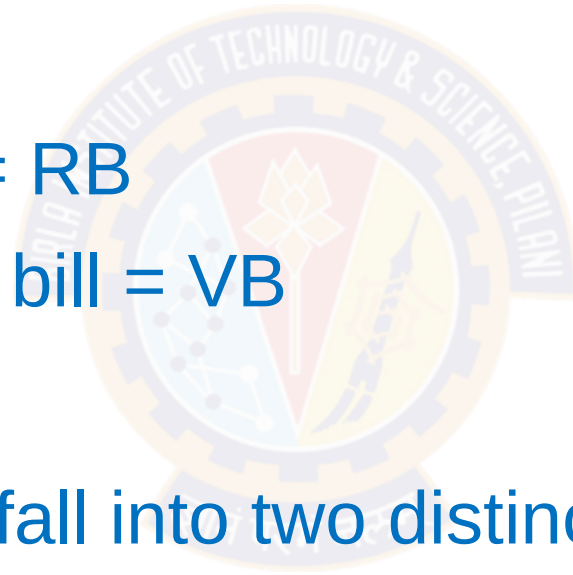
Using the Penn Tagset

- The/DT grand/JJ jury/NN commented/VBD on/IN a/DT number/NN of/IN other/JJ topics/NNS ./.
- Prepositions and subordinating conjunctions marked IN (“although/IN I/PRP..”)
- Except the preposition “to” is just marked “TO”.



POS Tagging

- Words often have more than one POS: back
 - The **back** door = JJ
 - On my **back** = NN
 - Win the voters **back** = RB
 - Promised to **back** the bill = VB
- POS-tagging algorithms fall into two distinctive groups:
 1. *Rule-Based POS Taggers Ex: Brill's tagger*
 2. *Stochastic POS Taggers Ex: Hidden Markov models*



NLTK POS Tagging Lab Demo

There are different open source POS taggers available

- Stanford PoS Tagger python bind- Java based, but can be used in python but difficult to install
- Flair - this is probably the most precise POS tagger available for python. It is almost 98% accurate.
- These two are the best ones.
- NLTK implementation to be very precise, around 97% and its quite fast.